

Agentic AI Readiness

What agentic AI is and why it determines whether this store gets recommended; how much of the required architecture is already built; what is blocking the rest; and the task list to finish it.

DATE	BUILD	APPLIES TO	COMPLETION
10 August 2026	main @ 704d1e6	Solana · BBQ · OKO	60%
1. What agentic AI actually means 2. Why it matters for this store 3. The four layers of readiness 4. Completion rate 5. What is already delivered 6. Blockers 7. Delivery plan 8. Task list 9. Testing & basis of assessment			

1. What agentic AI actually means

An ordinary AI assistant *answers*. An **agent acts** — it goes out, fetches pages, compares options and reports back with a recommendation. Increasingly it can complete the purchase too.

SOMEONE ASKS...	OLD BEHAVIOUR	AGENT BEHAVIOUR
"Find me a 4-burner built-in gas grill under \$3,000 that ships free"	Ten blue links. The shopper clicks, browses, filters, decides.	Visits candidate stores itself, reads specs and prices, filters, and returns two or three specific products — often with a buy button.

That is the shift. The shopper no longer visits ten sites. **An agent visits them on the shopper's behalf and returns a shortlist.** Being on that shortlist is the entire game.

The critical detail: most agents do not run JavaScript. A human browser downloads a page, runs its code, and assembles what you see. Most automated readers do not — they read the raw HTML the server sends and nothing more. If products are drawn in by JavaScript *after* the page arrives, the agent sees an empty page and moves on. It does not report an error. It simply finds nothing and recommends someone else.

2. Why it matters for this store

This catalogue is what agents get asked about Grills, fireplaces and outdoor kitchens are high-consideration, spec-heavy, \$1,500–\$5,000 purchases. Nobody impulse-buys a built-in grill. This is precisely the research-heavy category shoppers now delegate to an assistant.	Invisible is not neutral — it is lost In search you compete on <i>ranking</i>: position ten still exists. With an agent you are either machine-readable or absent from the shortlist entirely. There is no page two.
The same work pays for itself in ordinary SEO Server-rendered products and structured data are standard technical SEO. They improve normal Google results whether or not a single AI agent ever visits. Downside risk is close to zero.	Three storefronts, one codebase Solana, BBQ Grill Outlet and Outdoor Kitchen Outlet share the same code. Every improvement lands on all three at once — and so does every mistake, which is why brand-specific settings need care.

3. The four layers of readiness

"Agentic ready" is not one feature. It is four questions, in order. Each only matters if the one before it is already true — there is no point exposing a catalogue interface to an agent that was never allowed to fetch the site.

1 Reach — is the agent allowed in?

Crawler permissions and firewall rules. One misconfiguration can make the store invisible to AI systems. Also a policy question: some crawlers gather content to train models rather than send traffic.

2 Read — can it understand the page without running JavaScript?

Products, prices and availability present in the raw HTML, plus *structured data* — a standardised machine-readable description embedded in each page. **Nearly all the value sits here.**

3 Query — can it ask, instead of scraping?

A documented interface where an agent asks "4-burner built-in, under \$3,000, in stock" and gets a clean answer rather than downloading and guessing at pages. Faster, cheaper, far more accurate — the genuine competitive differentiator.

4 Transact — can it complete the purchase?

Emerging standards for agent-initiated checkout. Early, unsettled, and a poor fit for the current payment provider. A watching brief, not a project.

4. Completion rate

Scored against the twenty discrete deliverables in the agreed specification — one point for delivered, half for partial:

12 DELIVERED		8 NOT STARTED
20 deliverables	12 weighted score	60% complete

Progress since the specification was received. The build satisfied 8 of the 20 deliverables when this assessment began. Phase 1 has since closed four more — the consolidated context file, itemised specifications, semantic landmarks and aria-labels on product pages — taking it from **40% to 60%**.

A separate defect was found and fixed in the same pass: the storefront rendered a **completely blank page** with JavaScript disabled. The markup was present in the HTML, but React streamed it into a hidden element that only client-side JavaScript revealed — so requirement 2.7 was passing on a technicality. It is now genuinely satisfied on all three brands.

The remainder is not spread evenly. **Almost all of it sits in one requirement — the API and MCP layer.** That is a single coherent project rather than a long tail of small fixes, which makes it straightforward to schedule, and it is also the only part that is genuinely differentiating rather than table stakes.

\$	DELIVERABLE	STATUS	NOTE
1.1	Machine-readable site index (llms.txt)	Delivered	All three brands
1.2	Consolidated context file (llms-full.txt)	Delivered	Scoped for retail — \$6.3
1.3	Automatic regeneration on content change	Delivered	Exceeds requirement
2.1	Structured data on product pages	Delivered	
2.2	Structured data on FAQ content	Delivered	
2.3	Exact pricing and currency typed	Delivered	
2.4	Stock availability typed	Delivered	
2.5	Specifications typed	Delivered	24 attributes per product
2.6	Return policies typed	Blocked	Needs published terms — \$6.1
2.7	Not hidden behind client-side JavaScript	Delivered	Completed this week
3.1	Public rate-limited read endpoints	Not started	
3.2	OpenAPI specification	Not started	Path challenged — \$6.3
3.3	MCP support	Not started	
3.4	Machine-callable actions (cart, holds)	Not started	Scope challenged — \$6.3
4.1	AI user-agents not blocked	Delivered	22 agents named explicitly
4.2	Firewall / WAF audit	Blocked	No Cloudflare exists — \$6.2
4.3	Rate limiting (429 + Retry-After)	Not started	None exists — \$6.2
5.1	Live pricing and stock	Conflict	Architectural — \$6.2
5.2	Semantic HTML	Delivered	5 landmarks per product page
5.3	Meaningful aria-labels	Delivered	All landmarks labelled

5. What is already delivered

All three storefronts were run and inspected exactly as a non-JavaScript agent receives them.

The headline result

Before this work, **every product listing page returned zero products** to any reader that does not execute JavaScript — the precise failure described in §1. Resolved on all three brands.

PAGE TYPE	BEFORE	SOLANA	BBQ	OKO
Category page	0	30	30	30
Brand page	0	30	30	30
Main-nav page	0	30	30	30
Search results	0	30	30	30
Product page	11	11	11	11

Machine-readable descriptions

PAGE	BEFORE	NOW
Homepage	none	Store identity, site identity, search endpoint
Category / brand page	none	+ breadcrumb trail, list of 30 products with prices
Product page	basic	+ barcode, part number, condition, price validity, breadcrumbs, FAQs

Layers 1 and 2 are complete across all three brands. Layer 3 has not been started. Layer 4 is deliberately deferred.

6. Blockers

6.1 — Waiting on the client

ITEM	WHY IT BLOCKS
Published shipping & returns terms Free-shipping threshold, return window, restocking fees, who pays return postage	The one structured-data field we cannot fill. Terms that contradict the storefront or product feed are penalised by shopping platforms, so estimating is worse than leaving it blank. Blocks 2.6 / task T-09.
Decision: allow AI training crawlers?	All 22 currently allowed. Allowing means product copy and photography may train models with no traffic in return. Search and training crawlers can be separated, so "visible in AI answers but not used for training" is available. One-line change.
Decision: publish a public catalogue interface?	It makes the store natively usable by assistants — and is equally consumable by competitors. Gates the whole of Layer 3.
Hosting firewall check	Confirm whether bot protection is enabled on each of the three projects. Blocks 4.2 / task T-04.
Brand social accounts	BBQ and OKO still publish Solana's Facebook and Pinterest in their machine-readable identity, telling agents all three sites are one business. Fixed in code; must be cleared in hosting settings. Task T-05.

6.2 — Technical constraints

Live pricing and stock conflicts with the caching architecture. Six independent cache layers sit between a backend price change and what an agent sees — from 60 seconds up to 24 hours. This caching is deliberate and was introduced for page-speed reasons; taking the requirement literally means unwinding it and accepting materially slower pages on all three sites.

Proposed resolution: cached pages for humans, live data on the API for agents. Pages stay fast and publish a price-validity timestamp so an agent knows the freshness window; anything needing real-time stock calls the Layer 3 endpoint, which reads through to the backend.

Consequence: the live-data requirement depends on the API requirement. The original plan treats them as independent; they are not.

There is no rate limiting of any kind. Not CAPTCHA in place of a 429 — nothing at all, on any endpoint. Tolerable today because nothing valuable is exposed; not tolerable the moment a public catalogue interface exists. **Rate limiting must ship with the API, not after it.**

There is no Cloudflare or WAF in front of these sites. The platform is Vercel; the only Cloudflare reference is an analytics script. The requested audit is a no-op *unless* Vercel's own bot protection is enabled — not visible from the code, and must be checked in the dashboard. If it is on, it can silently block AI crawlers and undo the crawler-permission work already completed.

Semantic markup is weakest where it matters most. Across the storefront it is healthy — 133 `<section>`, 34 `<nav>`, 27 `<article>`, 171 `aria-label` attributes. But the three product-page components contain **zero** semantic elements and **zero** aria-labels. Product pages are the most important page type for a shopping agent. Contained fix.

6.3 — Points we are challenging in the specification

A full product-catalogue text dump is not workable at our scale. The cited guidance comes from documentation vendors, where a whole corpus is a few hundred kilobytes. Our catalogue exceeds 6,000 products; a complete dump would run to tens of megabytes — unfetchable, and too large for any context window. **Proposed instead:** a scoped file carrying policies, shipping terms, warranty, category structure and brands, pointing at the existing product feed for the catalogue. Same standard, sized for retail. Roughly 50 KB.

The proposed OpenAPI path is not a standard. It is not a registered well-known URI and agents do not look there automatically. We will publish at the conventional path, mirror at the requested one, and make the specification discoverable by explicit reference — but it should not be recorded as a standard, or a later reader will assume discovery happens by itself.

Cart creation and inventory holds are a much larger project than the read endpoints they are grouped with. Read-only access is contained and low risk. Letting agents create carts and hold stock introduces authentication, abuse prevention and inventory-locking design, and carries direct revenue risk. Recommend splitting into separate phases so the second does not inherit the first's estimate.

7. Delivery plan

Phase 0 — Unblock

IMMEDIATE · NO ENGINEERING TIME · CLIENT INPUT REQUIRED

Collect the five items in §6.1. Two of them gate work that is otherwise ready to start, so this should not wait for sprint planning.

Phase 1 — Close the quick gaps

ONE SPRINT · LOW RISK · NO NEW DEPENDENCIES

Scoped consolidated context file; semantic HTML and aria-labels on product pages; specifications itemised; shipping and returns published once Phase 0 delivers them; structured data on the category index; product-feed audit. **Now largely delivered** — only the shipping and returns copy remains, which is blocked on Phase 0. Took completion from 40% to 60%.

Phase 2 — API foundation

TWO SPRINTS · THE BULK OF THE REMAINING WORK

Rate limiting first, then read-only catalogue, search and inventory endpoints; a live read-through path so stock is accurate without touching page caching; OpenAPI published and referenced; crawler-permission carve-out. Resolves the live-data conflict as a by-product. Reaches roughly 90%.

Phase 3 — MCP connector

ONE SPRINT · SMALL ONCE PHASE 2 EXISTS

A Model Context Protocol server over the Phase 2 endpoints so assistants query the catalogue natively rather than browsing it. The genuinely differentiating deliverable.

Phase 4 — Transactional actions

SEPARATE PROJECT · SCOPE AND PRICE INDEPENDENTLY

Agent-callable cart creation and availability holds, plus evaluation of the emerging agent-payment protocols. Deliberately excluded from the estimate above.

8. Task list

ID	TASK	PHASE	SIZE	DEPENDS ON / BLOCKED BY
T-01	Obtain published shipping & returns terms	0	—	Client
T-02	Decision on AI training-crawler policy	0	—	Client
T-03	Decision on public catalogue interface	0	—	Client
T-04	Verify hosting bot-protection settings, all 3 projects	0	Small	Dashboard access
T-05	Clear cross-brand social accounts in hosting env	0	Small	Dashboard access
T-06	Build scoped consolidated context file	1	Small	Agreement on §6.3
T-07	Semantic HTML + aria-labels on product pages (×3 themes)	1	Small	—
T-08	Itemise product specifications in structured data	1	Small	—
T-09	Publish shipping & return policy in structured data	1	Small	T-01
T-10	Structured data on the category index page	1	Small	—
T-11	Audit product feed for barcodes, stock, shipping	1	Small	—
T-12	Rate limiting with 429 + Retry-After across API routes	2	Medium	Must precede T-13
T-13	Read-only catalogue + search endpoints	2	Medium	T-03, T-12

T-14	Live read-through inventory endpoint	2	Medium	T-13
T-15	OpenAPI specification, published and referenced	2	Small	T-13
T-16	Crawler-permission carve-out for the API paths	2	Small	T-13
T-17	Price-validity timestamps to bound cached staleness	2	Small	—
T-18	MCP server over the catalogue API	3	Medium	T-13, T-15
T-19	Agent test pass — verify with real assistants	3	Small	T-18
T-20	Evaluate agent-payment protocols	4	Small	Payment provider
T-21	Agent-callable cart creation & availability holds	4	Large	T-13, security review

9. Testing & basis of assessment

A test harness already exists and produced every figure in this document: each page is fetched exactly as a non-JavaScript agent receives it, and product links, prices and structured data are counted. It runs against any of the three brands in seconds and is suitable as a release gate. At Phase 3 we would add validation of every structured-data block against the schema.org validator, plus live query tests against the MCP connector using real assistants.

Basis. All figures measured on local development builds of all three brands on 10 August 2026 against `main @ 704d1e6` . Development and production rendering can differ; confirm on the live sites before treating this as a release record. Each brand is a separate deployment and all three must be released for completed work to take effect. The Layer 1 and 2 work rests on long-established web standards and is low risk; the Layer 4 landscape is young and moves quickly, so re-check before committing budget. Engineering detail is maintained alongside the code at `docs/agentic-ai-readiness.md` .